Certified Coherent-Collar Partition Optimization

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Abstract

We optimize the complete finite-time coherent-staple conditional estimate, including its exact $SU(2)$ Haar denominator and Poisson-image prefactors. A unique stationarity equation replaces the earlier unproved numerical-search optimality. Re-optimizing the local $K + R$ radius gives a nonempty conditional coherent-channel budget inside the certified nonlinear window. Collar occurrence for the complete measure remains open.

Let $v(\delta)$ be normalized Haar mass of an $SU(2)$ geodesic ball. For m staples in a common η-ball, heat time t, and target curvature ρ, Stage 15 gives

$$\mathbb{P}\{d(I, VQ_j) \geq \rho \mid \text{coherent collar}\} \leq \exp L_{t,\rho,\eta}(\delta),$$

where

$$L_{t,\rho,\eta}(\delta) = -\log v(\delta) + \log \frac{\rho}{\sin \rho} + (m+1) \log c_t - \frac{\rho^2 - m(\eta+\delta)^2}{2t},$$

and $0 < \delta < \rho/\sqrt{m} - \eta$.

Theorem 1 (Unique denominator optimizer)**.** *On every admissible interval containing its stationary point, $L_{t,\rho,\eta}$ has a unique global interior minimum δ_*, determined by*

$$\frac{v'(\delta_*)}{v(\delta_*)} = \frac{m(\eta+\delta_*)}{t}.$$

The optimizer is independent of ρ.

Proof. Differentiation gives

$$L'(\delta) = -\frac{v'(\delta)}{v(\delta)} + \frac{m(\eta+\delta)}{t}.$$

The radial density $\sin^2 \delta$ is strictly log-concave on the local interval. Its primitive $v(\delta)$ is log-concave, so v'/v is decreasing; the right side of the stationarity equation is strictly increasing. Near zero $v'/v \sim 3/\delta$, and hence a sign change gives exactly one root. The derivative changes from negative to positive there. Terms containing ρ are constant in δ, proving the final assertion. □

The executable certificate brackets the root with 120 bisections and records both sides of the equation. At $t = 3 \cdot 10^{-6}$ and $m = 6$, it treats both the ideal $\eta = 0$ envelope and the declared $\eta = 10^{-3}$ collar.

Theorem 2 (Conditional coherent-channel closure)**.** *Let $q = 0.002430277501841998$ be the Stage 25 weighted target. There exists an explicit local radius r_*, printed in the accompanying calibration, such that*

$$e^{\ell(r_*)}\, q_{\mathrm{ref}}(r_*) + q_{\mathrm{coh}}(2r_*, 10^{-3}) < q, \qquad r_* < 0.011453785959974176.$$

Here q_{ref} and ℓ are the certified local-reference complement and comparison loss, while q_{coh} is the optimized finite-time bound above.

Proof. All three functions are evaluated by the prior exact/certified modules. The calibration first brackets the crossing in the sharp nonlinear window and then uses 120 bisections, moving into the strict feasible side. The tests independently verify the stationarity residual, local minimality, crossing, strict budget, and radius inclusion. □

Remark 1 (Logical scope)**.** *The second theorem is conditional on collar coherence. It proves neither the probability of the coherence event nor the incoherent complement numerator. It also does not identify the abstract $K + R$ split with the complete nonlinear finite-volume kernel. Those missing statements are exactly the target of the next stage.*